

A S PRAJWAL BHAT

Bengaluru, Karnataka, India | prajwalbhat113@gmail.com
linkedin.com/in/a-s-prajwal-bhat-aa3717296 | github.com/AS-Prajwal-Bhat

SUMMARY

Final-year B.E. Electronics and Communication Engineering student (CGPA 8.5/10, zero backlogs) seeking an FPGA internship in RTL design, simulation and verification. Hands-on with Verilog RTL and FSM design, testbench-based simulation in Vivado and EDA Playground, and a from-scratch HDL-to-Verilog toolchain (CEDA). Trained in RTL-to-GDSII flow with Cadence tools; NPTEL Digital Design with Verilog (Elite + Silver). ESP32 hackathon winner with embedded firmware experience (ESP32, Arduino, 8051).

EDUCATION

Bangalore Institute of Technology — B.E. Electronics & Communication Engg. (CGPA 8.5/10, zero backlogs) *Expected 06/2027*

TECHNICAL SKILLS

HDL & Digital Design: Verilog (RTL design, FSM modeling, testbenches, simulation), digital logic design, combinational and sequential circuits, arbiters and arithmetic units

Tools & EDA: Xilinx Vivado, EDA Playground, Cadence tools (RTL-to-GDSII training), Git/GitHub, Linux/Ubuntu, basic MATLAB

Embedded Systems: ESP32, Arduino, 8051, sensor interfacing, real-time firmware, hardware-software integration

Programming: C, C++, Python, basic SQL, basic Java; Data Structures & Algorithms, OOP, Discrete Structures

CORE PROJECTS — DIGITAL DESIGN & FPGA

Vedic Multiplier in Verilog — Verilog, Vivado / EDA Playground

- Implemented a Vedic multiplier in Verilog; wrote testbenches and verified output against a standard multiplier baseline.

Round Robin Arbiter in Verilog — Verilog, Vivado / EDA Playground

- Designed and simulated an arbiter giving fair, cyclic access to a shared resource among multiple requesters.

Traffic Light Controller — Verilog, FSM Design

- Designed an FSM-based controller with multi-phase state transitions and signal timing logic, verified in simulation.

CEDA — Custom Electronic Design Automation Toolchain — C, Python, Tkinter

- Architected a C-based compiler translating a custom HDL to synthesizable Verilog, with a Python/Tkinter GUI front-end, covering lexing, parsing, semantic analysis and code generation.
- Debugged sensitivity-list generation, dropped module arguments and wire/reg inference errors, building a working grasp of how Verilog maps to hardware.

OTHER PROJECTS — EMBEDDED SYSTEMS

- RoboSentinel (Intelligent Surveillance System):** embedded control logic and real-time decision-making firmware; full hardware-software integration, testing and debugging.
- IoT-Based Smart Environment Monitoring:** real-time pipeline combining hardware sensors with Python data processing and structured anomaly logging.

INTERSHIPS

Internet of Things (IoT) Intern — Emertxe (online)

10 Aug 2026 – 16 Sep 2026

- Completed an online IoT internship with the project “Edge AI Based Smart EV Charging Station Optimizer”.

Student Internship Program 3.0 — IEEE CASS Bangalore Chapter

Oct 2025 – Dec 2025

CERTIFICATIONS & AWARDS

- Winner — IoT & Embedded Systems with ESP32 Hackathon** (Exovian Robotics @ BIT, Oct 2025)
- RTL to GDSII using Cadence Tools** — 5-day industry training, VLSIMINDS @ BIT ECE (Apr 2026)
- NPTEL Digital Design with Verilog** (IIT Guwahati) — Elite + Silver, 75%
- NIELIT Calicut: VLSI for Beginners** — Chips to Startup (C2S) Programme, Sept 2026
- NPTEL CMOS Digital VLSI Design** (IIT Roorkee) — Elite
- NPTEL Joy of Computing using Python** (IIT Madras) — Elite + Silver, 86%, Top 5% Topper
- NPTEL Introduction to IoT** (IIT Kharagpur) — Elite + Silver, 81%; **Problem Solving Through Programming in C** — Elite

LEADERSHIP EXPERIENCE

Secretary — IEEE CEDA Society & IEEE CAS Society, BIT Student Branch

Dec 2025 – Present

- Coordinating technical design initiatives alongside branch-wide event leadership.

General Secretary — IEEE Student Branch, Bangalore Institute of Technology

Aug 2025 – Present

- Organized 4–5 technical workshops (TCL, Linux, VLSI) and hands-on competitions (Arduino, Verilog) with a 100% event execution success rate.
- Led a national-level hackathon with around 80 participants from across India, managing scheduling across volunteer teams.

Placement Coordinator — Bangalore Institute of Technology

Associate Design Lead — IEEE CEDA Society & IEEE CAS Society, BIT Student Branch

Mar 2025 – Aug 2025

- Led a college-wide Verilog Design Competition; helped run the “SmartEco+ Design” Hackathon (₹30,000 prize pool).